

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

HSC ASSESSMENT TASK 3

JUNE 2006

MATHEMATICS
EXTENSION 1

Time Allowed: 70 Minutes

Instructions:

- Write your name and class at the top of each page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new page
- Standard integrals can be found on the last page.

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/10	/10	/10	/10	/10	/12	/62

Question 1

- (a) Find the exact value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ 1
- (b) Differentiate the following
- i. $y = e^{x^2}$ 1
- ii. $y = \sin^{-1} 3x$ 2
- iii. $y = \log_e \frac{x^2}{x-1}$ 2
- (c) Simplify $\log_a x^3 \div \log_a \sqrt{x}$ 2
- (d) Find the general solution of $2\cos x = -1$ 2

Question 2 (Begin on the next page)

- (a) Find
- i. $\int \frac{x}{x^2+1} dx$ 1
- ii. $\int 2x^2 e^{x^3} dx$ 1
- (b) Find k if $x^{k+3} = e^{7\ln x}$ where $x > 0$ 2
- (c) Consider $f(x) = \sin^{-1}(2x+1)$
- i. Find the domain of $y = f(x)$ 1
- ii. Find the inverse function $y = f^{-1}(x)$ 2
- iii. State the domain and range of $y = f^{-1}(x)$ 2
- iv. Sketch $y = f^{-1}(x)$ 1

Question 3 (Begin on the next page)

- (a) Consider the function $y = x \ln x$ where $x > 0$.
- i. Find the stationary point and determine its nature. 2
 - ii. Show that the curve is always concave upwards. 1
 - iii. Find $\lim_{x \rightarrow 0} x \ln x$. 1
 - iv. Where does the curve cut the x axis? 1
 - v. Sketch the curve showing all important features. 2
- (b) Find $\int \frac{dx}{\sqrt{16 - 25x^2}}$. 3

Question 4 (Begin on the next page)

- (a) During the early summer months the rate of increase of the population, P , of fruit flies is proportional to the excess of the population over 3000 so $\frac{dP}{dt} = k(P - 3000)$ where t is in months and k is a constant.
- At the beginning of summer the population is 4000 and 1 month later it is 10 000.
- i. Show that $P = 3000 + Ae^{kt}$ is a solution of the differential equation, 1
where A is a constant.
 - ii. Find the value of A and the exact value of k . 2
 - iii. Find to the nearest 100, the population after $2\frac{1}{2}$ months. 1
 - iv. After how many weeks does the population reach $\frac{1}{2}$ million? 2
- (b) A spherical balloon is being inflated and its volume increases at the constant rate of 50 mm^3 per second.
- At what rate is its surface area increasing when the radius is 20mm? 4

Question 5 (Begin on the next page)

(a) i. Show that $\int_0^{\frac{\pi}{4}} \cos^2 \theta d\theta = \frac{\pi + 2}{8}$. 3

ii. Hence, using the substitution $x = 2\sin\theta$ or otherwise, 3
evaluate $\int_0^{\sqrt{2}} \sqrt{4 - x^2} dx$

(b) Use the substitution $x = u^2, u > 0$ to find the exact value of 4

$$\int_1^3 \frac{1}{(x+1)\sqrt{x}} dx$$

Question 6 (Begin on the next page)

i. Solve the equation $x^4 + x^2 - 1 = 0$, giving answers correct to 2
two decimal places.

ii. On the same axes, draw the graphs of $y = \tan^{-1}x$ and $y = \cos^{-1}x$, 2
showing all important features. Mark the point P where the
curves intersect.

iii Let $\tan^{-1}x = \alpha$. Find an expression for $\cos \alpha$ in terms of x . 1

iv. Show that, if $\tan^{-1}x = \cos^{-1}x$ then $x^4 + x^2 - 1 = 0$. 2

v. State the coordinates of P. 1

vi. Find the area enclosed by the curves and the y axis from 4
 $y = \frac{\pi}{2}$ to $y = 0$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

$$(a) \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\tan^{-1}\frac{1}{\sqrt{3}} \\ = \underline{\underline{-\frac{\pi}{6}}}$$

$$(b) \text{ i. } \frac{dy}{dx} = \underline{\underline{2xe^{x^2}}}$$

$$\text{ii. } \frac{dy}{dx} = \underline{\underline{\frac{3}{\sqrt{1-9x^2}}}}$$

$$\text{iii. } y = \log_e x^2 - \log_e (x-1) \\ y = 2 \log_e x - \log_e (x-1) \\ \frac{dy}{dx} = \underline{\underline{\frac{2}{x} - \frac{1}{x-1}}}$$

$$(c) \frac{3 \log_a x}{\frac{1}{2} \log_a x} = \underline{\underline{6}}$$

$$(d) \cos x = -\frac{1}{2}$$

$$\underline{\underline{x = 2n\pi \pm \frac{2\pi}{3} \text{ where } n \text{ is an integer}}}$$

Question 2

$$(a) \text{ i. } \frac{1}{2} \int \frac{2x}{x^2+1} dx = \underline{\underline{\frac{1}{2} \ln(x^2+1) + c}}$$

$$\text{ii. } \frac{2}{3} \int 3x^2 e^{x^3} dx = \underline{\underline{\frac{2}{3} e^{x^3} + c}}$$

$$(b) \begin{aligned} x^{k+3} &= e^{7 \ln x} \\ x^{k+3} &= e^{\ln x^7} \\ x^{k+3} &= x^7 \end{aligned}$$

$$(c) f(x) = \sin^{-1}(2x+1)$$

$$\text{i. } -1 \leq 2x+1 \leq 1 \\ -2 \leq 2x \leq 0$$

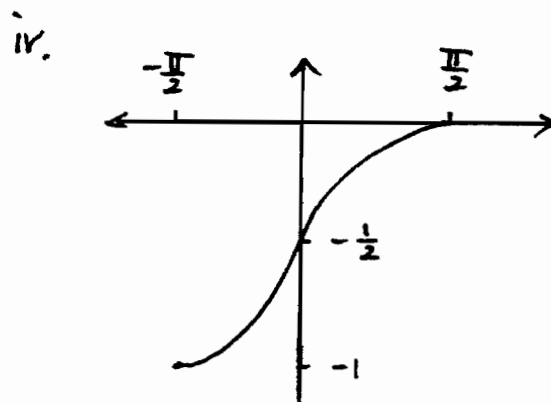
$$D: \underline{\underline{-1 \leq x \leq 0}}$$

$$\text{ii. } x = \sin^{-1}(2y+1) \\ \sin x = 2y+1$$

$$\therefore \underline{\underline{y = \frac{1}{2}(\sin x - 1)}}$$

$$\text{iii. } D: \underline{\underline{-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}}}$$

$$R: \underline{\underline{-1 \leq y \leq 0}}$$

Question 3

$$(a) y = x \ln x$$

$$\text{i. } \frac{dy}{dx} = x \cdot \frac{1}{x} + \ln x \cdot 1 \\ = 1 + \ln x$$

$$\text{Stat pts: } \frac{dy}{dx} = 0$$

$$0 = 1 + \ln x$$

$$\ln x = -1$$

$$x = \frac{1}{e}, y = -\frac{1}{e}$$

$$\frac{d^2y}{dx^2} = \frac{1}{x}$$

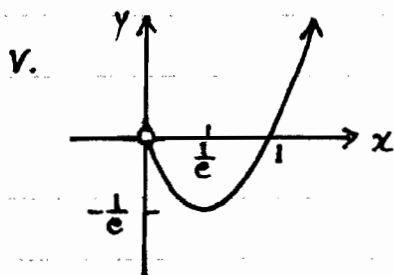
$$(1 - 1)$$

ii. $\frac{d^2y}{dx^2} = \frac{1}{x} > 0$ for all $x > 0$

$\therefore$ the curve is always concave up

iii. $\lim_{x \rightarrow 0} x \ln x = \underline{\underline{0}}$

iv. $\underline{\underline{(1, 0)}}$



$$\begin{aligned}
 \text{(b)} \int \frac{dx}{\sqrt{16-25x^2}} &= \int \frac{dx}{\sqrt{25\left(\frac{16}{25}-x^2\right)}} \\
 &= \frac{1}{5} \int \frac{dx}{\sqrt{\frac{16}{25}-x^2}} \\
 &= \frac{1}{5} \sin^{-1} \frac{x}{\frac{4}{5}} + C \\
 &= \underline{\underline{\frac{1}{5} \sin^{-1} \frac{5x}{4} + C}}
 \end{aligned}$$

Question 4

(a) i. $P = 3000 + Ae^{kt}$
 $\frac{dP}{dt} = kAe^{kt}$
 $= k(P - 3000)$
 since $P = 3000 + Ae^{kt}$

ii. when $t=0$, $P=4000$
 $4000 = 3000 + Ae^0$
 $\therefore \underline{\underline{A = 1000}}$

when $t=1$, $P=10\,000$
 $10\,000 = 3000 + 1000e^k$
 $7000 = 1000e^k$
 $e^k = 7$
 $\therefore \underline{\underline{k = \ln 7}}$

iii. when $t=2.5$, $P=?$
 $P = 3000 + 1000e^{\ln 7 \times 2.5}$
 $= 132\,641.8..$
 $= \underline{\underline{132\,600}}$

iv. $P=500\,000$, $t=?$
 $500\,000 = 3000 + 1000e^{(\ln 7)t}$
 $497\,000 = 1000e^{(\ln 7)t}$
 $497 = e^{(\ln 7)t}$
 $t = \frac{\ln 497}{\ln 7}$
 $= 3.19 \text{ months}$
 $\approx \underline{\underline{13 \text{ weeks}}}$

(b) $V = \frac{4}{3}\pi r^3$
 $S = 4\pi r^2$

$$\begin{aligned}
 \frac{dV}{dt} &= \frac{dV}{dr} \times \frac{dr}{dt} \\
 &= 4\pi r^2 \frac{dr}{dt} \\
 50 &= 4\pi \times 20^2 \times \frac{dr}{dt} \\
 \frac{dr}{dt} &= \frac{5}{160\pi}
 \end{aligned}$$

$$\begin{aligned}
 \frac{dS}{dt} &= \frac{dS}{dr} \times \frac{dr}{dt} \\
 &= 8\pi r \times \frac{dr}{dt} \\
 &= 8\pi \times 20 \times \frac{5}{160\pi}
 \end{aligned}$$

Question 5

a) i. $\int_0^{\frac{\pi}{4}} \cos^2 \theta \, d\theta$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 2\theta + 1 \, d\theta$$

$$= \frac{1}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left[\left(\frac{1}{2} \sin 2 \cdot \frac{\pi}{4} + \frac{\pi}{4} \right) - \left(\frac{1}{2} \sin 0 + 0 \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} + \frac{1}{2} \right]$$

$$= \frac{\pi + 2}{8}$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\cos^2 \theta = \frac{1}{2} (\cos 2\theta + 1)$$

ii. $\int_0^{\sqrt{2}} \sqrt{4-x^2} \, dx$

$$= \int_0^{\frac{\pi}{4}} \frac{\sqrt{4-4\sin^2 \theta} \times 2 \cos \theta \, d\theta}{4(1-\sin^2 \theta)}$$

$$= \int_0^{\frac{\pi}{4}} \sqrt{4\cos^2 \theta} \times 2 \cos \theta \, d\theta$$

$$= 4 \int_0^{\frac{\pi}{4}} \cos^2 \theta \, d\theta$$

$$= 4 \times \frac{\pi + 2}{8}$$

$$= \frac{\pi + 2}{2}$$

$$x = 2 \sin \theta$$

$$\frac{dx}{d\theta} = 2 \cos \theta$$

$$dx = 2 \cos \theta \, d\theta$$

$$x = \sqrt{2} : \sqrt{2} = 2 \sin \theta$$

$$\frac{1}{\sqrt{2}} = \sin \theta$$

$$\theta = \frac{\pi}{4}$$

$$x = 0 : 0 = 2 \sin \theta$$

$$\theta = 0$$

(b) $\int_1^3 \frac{1}{(1+x)\sqrt{x}} \, dx$

$$= \int_1^{\sqrt{3}} \frac{1}{(1+u^2)u} \times 2u \, du$$

$$= 2 \int_1^{\sqrt{3}} \frac{1}{1+u^2} \, du$$

$$= 2 \left[\tan^{-1} u \right]_1^{\sqrt{3}}$$

$$= 2 \left[\tan^{-1} \sqrt{3} - \tan^{-1} 1 \right]$$

$$= 2 \left[\frac{\pi}{3} - \frac{\pi}{4} \right]$$

$$= \frac{\pi}{6}$$

$$x = u^2$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u \, du$$

$$x = 3 : 3 = u^2$$

$$u = \sqrt{3}$$

$$x = 1 : 1 = u^2$$

$$u = 1$$

Question 6

i. $x^4 + x^2 - 1 = 0$

let $x^2 = a$

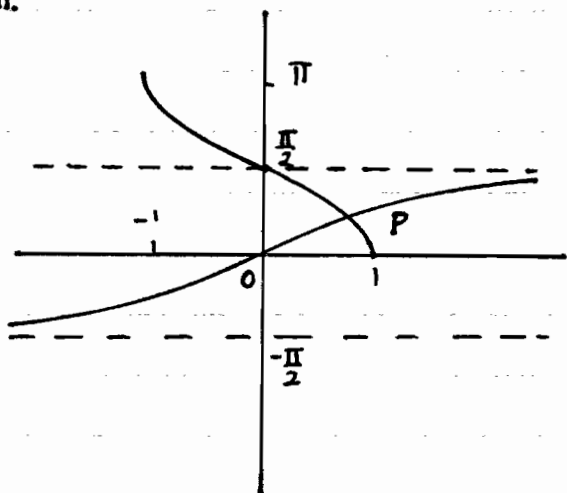
$a^2 + a - 1 = 0$

$a = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot (-1)}}{2}$

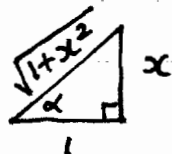
$x^2 = \frac{-1 \pm \sqrt{5}}{2}$

$\therefore \underline{\underline{x = \pm 0.79}}$

ii.



iii. let $\tan^{-1} x = \alpha$
 $\tan \alpha = x$



$\therefore \cos \alpha = \frac{1}{\sqrt{1+x^2}}$

iv. If $\tan^{-1} x = \cos^{-1} x$

$\cos^{-1} x = \alpha$

$\cos \alpha = x$

$\frac{1}{\sqrt{1+x^2}} = x$

$\frac{1}{1+x^2} = x^2$

v. $P(0.79, 0.67)$

vi. $y = \tan^{-1} x$
 $x = \tan y$

$y = \cos^{-1} x$
 $x = \cos y$

$A = \int_{0.67}^{\frac{\pi}{2}} \cos y \, dy$
 $+ \int_0^{0.67} \tan y \, dy$

$= \int_{0.67}^{\frac{\pi}{2}} \cos y \, dy + \int_0^{0.67} \frac{\sin y}{\cos y} \, dy$

$= \left[\sin y \right]_{0.67}^{\frac{\pi}{2}} + \left[-\ln(\cos y) \right]_0^{0.67}$

$= \left[\sin \frac{\pi}{2} - \sin 0.67 \right] +$

$\left[(-\ln(\cos 0.67)) - (-\ln(\cos 0)) \right]$

$= \underline{\underline{0.62}} \quad (2 \text{ dp})$